

CERTIFICATE OF ANALYSIS

Product Name	GLYCYRRHIZA GLABRA (MULETHI) DRY EXTRACT - 20% BY HPLC		
Date of Analysis	18-01-2023	Date of Manufacturing	Jan-2023
Batch No.	LC-230240	Date of Expiry	Dec-2025
T.R. No.	HC/HE/F/2023/O0240		
Category	Intended for Cosmeceutical and Neutraceutical appl	Plant Part	Roots
Botanical name	<i>Glycyrrhiza glabra</i>	Standardization	Glycyrrhizic Acid

Parameters	Specifications	Result	Reference
Physical			
Description	Brown coloured powder	Complies	Visual
Identification	To comply with std	Complies	
Loss on drying	Not more than 6.0% w/w (dried at 105°C)	4.96 % w/w	USP <731>
Tapped bulk density	Between 0.40 g/ml and 0.80 g/ml	0.70g/ ml	USP <616>
Sieve Test (Passes through)			
40 mesh	Not less than 100% w/w	100% w/w	
60 mesh	Not less than 100% w/w	100% w/w	
80 mesh	Not less than 85% w/w	90.46% w/w	
Ash Content	Not more than 20.0% w/w	7.72% w/w	USP <561>
Water soluble (1% w/v solution in water)	Not less than 80.0% w/w	84.59% w/w	
Chemical Assay			
Glycyrrhizic acid Content (By HPLC)	Not less than 20% w/w	20.59%w/w	IN HOUSE
Others			
Residual pesticides	To comply as per USP	Complies	USP <561>
Total heavy metals	Not more than 20 ppm	Complies	USP<231> Method II
Lead	Not more than 3 ppm	Complies	ICP-OES, SLL/STP-II-006
Arsenic	Not more than 1 ppm	Complies	ICP-OES, SLL/STP-II-006
Cadmium	Not more than 1 ppm	Complies	ICP-OES, SLL/STP-II-006
Mercury	Not more than 0.1 ppm	Complies	ICP-OES, SLL/STP-II-006
Residual solvents	To comply as per USP	Complies	USP <467>

Manufacturing Unit:
Herbal Creations
(A unit of Kumaon Exports Pvt. Ltd.)
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Lic No.:10020012000643

CERTIFICATE OF ANALYSIS

Product Name	GLYCYRRHIZA GLABRA (MULETHI) DRY EXTRACT-20% BY GRAVIMETRY		
Date of Analysis	18-01-2023	Date of Manufacturing	Jan-2023
Batch No.	LC-230239	Date of Expiry	Dec-2025
T.R. No.	HC/HE/F/2023/O0239		
Category	Intended for Cosmeceutical and Neutraceutical appl	Plant Part	Roots
Botanical name	<i>Glycyrrhiza glabra</i>	Standardization	Glycyrrizic Acid

Parameters	Specifications	Result	Reference
Physical			
Description	Brown coloured powder	Complies	Visual
Identification	To comply with std	Complies	
Loss on drying	Not more than 6.0% w/w (dried at 105°C)	4.95 % w/w	USP <731>
Tapped bulk density	Between 0.40 g/ml and 0.80 g/ml	0.74 g/ ml	USP <616>
Sieve Test (Passes through)			
40 mesh	Not less than 100% w/w	100% w/w	
60 mesh	Not less than 100% w/w	100% w/w	
80 mesh	Not less than 85% w/w	89.76% w/w	
Ash Content	Not more than 20.0% w/w	8.84% w/w	USP <561>
Water soluble (1% w/v solution in water)	Not less than 80.0% w/w	83.73 % w/w	
Chemical Assay			
Content of Glycyrrhizic (By Gravimetry)	Not less than 20.0%	20.57%w/w	IN HOUSE
Others			
Total heavy metals	Not more than 20 ppm	Complies	USP<231> Method II
Lead	Not more than 3 ppm	Complies	ICP-OES ,SLL/STP-II-006
Arsenic	Not more than 1 ppm	Complies	ICP-OES ,SLL/STP-II-006
Cadmium	Not more than 1 ppm	Complies	ICP-OES ,SLL/STP-II-006
Mercury	Not more than 0.1 ppm	Complies	ICP-OES ,SLL/STP-II-006
Residual solvents	To comply as per USP	Complies	USP <467>
Residual pesticides	To comply as per USP	Complies	USP <561>

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